§5.1 The coefficients: Full Fourier series

$\left\{1, \cos\frac{\pi x}{L}, \sin\frac{\pi x}{L}, \cos\frac{2\pi x}{L}, \sin\frac{2\pi x}{L}, \cdots, \cos\frac{n\pi x}{L}, \sin\frac{n\pi x}{L}, \cdots\right\}$

Definition

The full Fourier series, or simply the Fourier series, of $\phi(x)$ on the interval $-l < x < l$, is defined as

$$\phi(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left(A_n \cos\frac{n\pi x}{l} + B_n \sin\frac{n\pi x}{l}\right), \tag{34}$$

where

$$\begin{aligned} A_n &= \frac{1}{l}\int_{-l}^{l} \phi(x)\cos\frac{n\pi x}{l}\,dx, \quad n = 0, 1, 2, \cdots, \\ B_n &= \frac{1}{l}\int_{-l}^{l} \phi(x)\sin\frac{n\pi x}{l}\,dx, \quad n = 1, 2, \cdots. \end{aligned} \tag{35}$$

§5.1 The coefficients: Full Fourier series

Example 5

Expand $\phi(x) = x, x \in [-l, l]$ in full Fourier series and sketch the graph.

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Solution: We need to evaluate

$$A_n = \frac{1}{l}\int_{-l}^{l} x \cos\frac{n\pi x}{l}\,dx, \quad n = 0, 1, 2, \cdots$$

and

$$B_n = \frac{1}{l}\int_{-l}^{l} x \sin\frac{n\pi x}{l}\,dx, \quad n = 1, 2, \cdots .$$

f is odd

$$\int_{-a}^{0} f(x)\,dx \overset{-x\to x}{=} \int_{a}^{0} f(-x)(-1)\,dx = \int_{a}^{0} f(x)\,dx = -\int_{0}^{a} f(x)\,dx$$

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and

$$B_n = \frac{1}{l}\int_{-l}^{l} x\sin\frac{n\pi x}{l}\,dx, \quad n = 1, 2, \cdots.$$

For each A_n, since the integrand $x\cos\frac{n\pi x}{l}$ is an odd function in the interval $(-l, l)$, $A_n = 0$.

If $f(x)$ is odd on $(-a, a)$

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$$

$$x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2l}{n\pi} \sin \frac{n\pi x}{l}$$

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Integration by parts

$$B_n = \frac{1}{l} \int_{-l}^{l} x \sin \frac{n\pi x}{l} dx = \left(\frac{-x}{n\pi} \cos \frac{n\pi x}{l} + \frac{1}{n^2\pi^2} \sin \frac{n\pi x}{l} \right) \Big|_{-l}^{l} = (-1)^{n+1} \frac{2l}{n\pi}.$$

§5.1 The coefficients: Full Fourier series

We plot the function $\phi(x) = x$ in $-3 = -l < x < l = 3$, and partial sums $S_m(x)$.

$\frac{2l}{\pi} \sin \frac{\pi x}{l}$

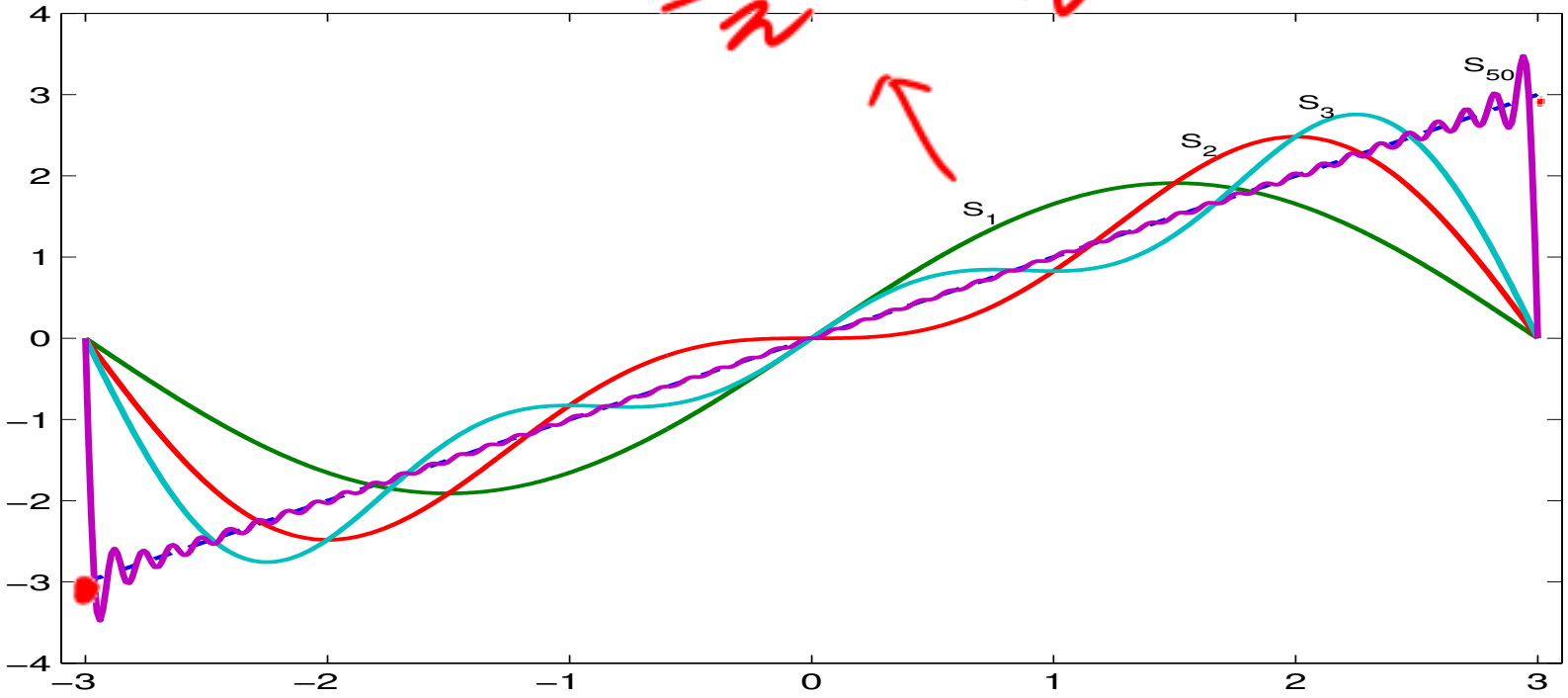

Figure: Fourier series of x (marked by dash-line) and S_1, S_2, S_3, S_{50}.

Example 6

Solve the problem

$$
\begin{aligned}
&u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \ t > 0, \\
&u(0,t) = u(l,t) = 0, \quad t > 0, \\
&u(x,0) = x, \quad u_t(x,0) = 0, \quad 0 < x < l,
\end{aligned}
\tag{36}
$$

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Solution: From (5), we know that $u(x,t)$ has an expansion

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}.$$

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Differentiating with respect to time yields

$$u_t(x,t) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} \left(-A_n \sin \frac{n\pi ct}{l} + B_n \cos \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}.$$

Example 6 (Cont'd)

Setting $t = 0$ in the expansion of $u_t(x, t)$, we have

$$0 = u_t(x, 0) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}.$$

so that all the $B_n = 0$.

$$\left\{ \sin \frac{n\pi x}{l} \right\}_{n=1}^{\infty}$$

linearly independent.

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This is exactly the series of Example 3.

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This is exactly the series of Example 3. Therefore, the complete solution is

$$u(x, t) = \frac{2l}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cos \frac{n\pi ct}{l} \sin \frac{n\pi x}{l}.$$

$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} + \cdots$

§5.1 The coefficients

Example A (not in the textbook)

Solve the problem

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l,\ t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = 0, \quad u_t(x,0) = 1, \quad 0 < x < l, \end{aligned} \tag{37}$$

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Setting $t = 0$ in the expansion of $u(x, t)$, we have

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Example 1 gives the Fourier sine expansion of function $f(x) = 1$ in $(0, l)$:

$$1 = \sum_{n=1}^{\infty} \frac{2\left[1 - (-1)^n\right]}{n\pi} \sin \frac{n\pi x}{l}.$$

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So we have B_n and the consequent solution

$$B_n = \frac{2l\left[1 - (-1)^n\right]}{n^2\pi^2 c},$$

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2l\left[1 - (-1)^n\right]}{n^2\pi^2 c} \sin \frac{n\pi ct}{l} \sin \frac{n\pi x}{l}.$$

§5.4 Completeness

We state the basic theorems about the convergence of Fourier series in §5.4.

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Three notions of convergence

(i) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ **pointwisely**, if for each $x \in (a, b)$,

$$\lim_{N\to\infty} S_N(x) = f(x).$$

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

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open interval

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(ii) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ **uniformly**, if

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(iii) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ in the L^2-**sense** (or the **mean-square sense**), if

$$\lim_{N\to\infty} \int_a^b |f(x) - S_N(x)|^2 dx = 0.$$

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Example 1. Pointwise $\not\Rightarrow$ Uniform

Let $f_n(x) = (1-x)x^{n-1} = x^{n-1} - x^n$ on the interval $(0,1)$. Check the convergence of the series $\sum_{n=1}^{\infty} f_n(x)$ to $f(x) = 1$.

Solution: The partial sums are

$$\lim_{N\to+\infty}(1-x^N) = 1$$

$$\sum_{n=1}^{N} f_n(x) = \sum_{1}^{N}(x^{n-1} - x^n) = 1 - x^N \to 1 \quad \text{as} \quad N \to \infty. \quad 0<x<1$$

This convergence is valid for each x. In words, the series converges pointwise to the function $f(x) = 1$. But the convergence is not uniform because $\max[1-(1-x^N)] = \max x^N = 1 \neq 0$ for every N. However, it does converge in mean-square since $\int_0^1 |x^N|^2 dx = \frac{1}{2N+1} \to 0$.

Theorem 2. Uniform Convergence

The Fourier (sine, cosine or full) series of $f(x)$, converges uniformly to $f(x)$ provided that

(i) for Fourier sine series: $f(x)$ and $f'(x)$ are continuous on $[0, l]$ and $f(0) = f(l) = 0$;

1

2

$f(x) = 1$, $f(x) = x$

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1

$f(x) = 1$ $\quad$ $g(x) = x$

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(ii) for Fourier cosine series: $f(x)$ and $f'(x)$ are continuous on $[0, l]$;

(iii) for full Fourier series: $f(x)$ and $f'(x)$ are continuous on $[-l, l]$ and $f(-l) = f(l)$.

1

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Remark: The Fourier cosine series does not require the conditions at the endpoints, as the even extension will lead to $f(-l) = f(l)$.

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Example: Examine the Fourier sine, cosine and Fourier full series of x that we discussed before.

Theorem 3. L^2-Convergence

The Fourier series converges to $f(x)$ in the mean-square sense in (a, b) provided only that $f(x)$ is any function for which

$$\int_a^b |f(x)|^2 dx$$

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Remark: The condition is weak, which means the Fourier series of a large class of functions converge in L^2-sense. Notice that all continuous functions are in L^2.

Theorem 4. Pointwise Convergence of Classical Fourier Series

(i) The classical Fourier series (full or sine or cosine) converges to $f(x)$ pointwisely on (a, b) provided that $f(x)$ is a continuous function on $[a, b]$, and $f'(x)$ is piecewise continuous on $[a, b]$.

① ②

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(ii) More generally, if f itself is only piecewise continuous on $[a, b]$ and $f'(x)$ is also piecewise continuous on $[a, b]$, then the Fourier series converges at every point $x \in (\cancel{-\infty}\, a, \cancel{\infty}\, b)$. Moreover, we have the convergence:

f may have

$$\sum_n A_n X_n(x) = \frac{1}{2}\big[f(x+) + f(x-)\big], \quad -\infty < x < \infty, \tag{38}$$

where $f(a\pm) = \lim_{x\to a^\pm} f(x)$. Here, $f(x)$ is the extension (periodic, odd periodic or even periodic) of f in the original interval, e.g., $(0, l)$ or $(-l, l)$.

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Thus, at a jump discontinuity, the series converges to the average of the limits from the right and from the left. In the case of the Fourier sine (or cosine) series on $(0, l)$, the extended function $f_{\text{ext}}(x)$ is the odd (or even) function of period $2l$. For the full series on $(-l, l)$, it is the periodic extension. The extension is piecewise continuous with a piecewise continuous derivative on $(-\infty, \infty)$.

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